

Lab Task – 2

Reinforcement Learning Laboratory

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Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

RL Demo: <https://jinglescode.github.io/reinforcement-learning-tic-tac-toe/>

Task 1: Identify the Reinforcement Learning Components

RL Component	Student Observation
Agent	The AI player is the learning agent. It observes the board, chooses where to place its mark, and updates its strategy based on the outcome of each game.
Environment	The 3x3 Tic-Tac-Toe board is the environment. It receives the agent's move, updates the board configuration, and signals whether the game has ended (win, loss, or draw).
State Representation	A state is the current pattern of X, O, and empty cells across the 9 board positions. Every move transitions the game to a new state.
Action Space	The action space is the set of empty cells the agent can mark on its turn. It shrinks as the game progresses and is empty once the board is full.
Reward Function	The agent gets a positive reward for a win, a negative reward for a loss, and a small/neutral reward for a draw. These rewards are propagated back to guide future move choices.
Learning Approach Used	The demo uses value-based Reinforcement Learning through self-play (an approach similar to TD-learning), where two agent instances repeatedly play each other and update a value function for board states.

Answers to the Analysis Questions

1. Who is the learning agent in this environment?

Ans: The learning agent is the AI player (marked O in the demo). It is the decision-maker that selects a cell to play on each of its turns and adjusts its behaviour based on the rewards it collects over many games.

2. What is considered as the environment?

Ans: The Tic-Tac-Toe board, together with the game rules, forms the environment. It is external to the agent, it changes state in response to the agent's actions, and it determines when the game ends and what the outcome is.

3. How is the game state represented?

Ans: The state is represented as the current configuration of the 3x3 grid, i.e. which of the 9 cells contain an X, an O, or are still empty. Each unique arrangement of the board corresponds to a distinct state that the agent can learn a value for.

4. What are the possible actions available to the agent?

Ans: At any turn, the available actions are the empty cells on the board. Choosing an action means placing the agent's mark in that cell. The number of legal actions decreases by one after every move made by either player.

5. When does the agent receive a positive reward?

Ans: The agent receives a positive reward when it wins the game, i.e. when it completes three of its marks in a row, column, or diagonal before the opponent does. Losing yields a negative reward and a draw yields a small/neutral reward.

6. Which Reinforcement Learning approach is used for learning?

Ans: The demo uses a value-based, model-free approach in which the agent learns a value function over board states through self-play, updating its estimates using a TD-style update rule after each game.

7. Is the agent trained using labelled data? Justify your answer.

Ans: No. The agent is not given labelled examples of "correct" moves. Instead, it learns purely from experience, playing many games against itself/another agent and updating its value estimates from the win/loss/draw reward signal it receives, which is the defining characteristic of reinforcement learning rather than supervised learning.

8. Explain how Reinforcement Learning differs from Supervised Learning using this example.

Ans: In Supervised Learning, the agent would need a dataset of board states each paired with the "correct" move, and it would learn to imitate those labels. In this Tic-Tac-Toe demo, no such labels exist; the agent instead takes actions, observes a delayed reward at the end of the game, and gradually improves its policy through trial and error. RL therefore learns from the consequences of its own actions, while Supervised Learning learns from a fixed set of correct answers.

Observation

The RL agent learns to play Tic-Tac-Toe by repeatedly interacting with the board. The board acts as the environment, the arrangement of marks on it is the state, and placing a mark in an empty cell is the action. Rather than being given labelled examples, the agent improves through self-play, using the win/loss/draw reward signal to gradually refine which moves lead to better outcomes.

Task 2: Effect of Training Episodes on Agent Behaviour

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	28%	58%	14%	Plays almost randomly, frequently misses obvious blocks, and loses most games.
500	42%	38%	20%	Starts blocking some opponent threats, occasionally takes the center, but still makes avoidable mistakes.
1,000	58%	20%	22%	Prefers the center and corners, blocks winning threats more consistently, and shows clearer strategy.
5,000	74%	8%	18%	Rarely loses, actively forces draws when it cannot win, and plays consistently strong moves.
10,000	85%	3%	12%	Plays near-optimally, almost never loses, and reliably forces a draw against strong opposition.

Behaviour Checklist Observed During Training

- Plays randomly — seen mainly at 100 episodes.
- Blocks opponent moves — emerging by 500, reliable by 1,000+.
- Chooses the center position — becomes a consistent opening preference from 500 episodes onward.
- Chooses corner positions — used strategically once training passes 1,000 episodes.
- Makes strategic moves — clearly visible from 1,000 episodes onward.
- Avoids losing — strong from 5,000 episodes onward.
- Forces a draw — common outcome at 5,000-10,000 episodes.
- Plays optimally — approached at 10,000 episodes.

Answers to the Analysis Questions

9. How does the quality of play change as the number of training episodes increases?

Ans: Play quality rises steadily with more training. Early on the agent moves almost randomly and loses often; as episodes increase it starts to block threats, prefers strong opening squares, and eventually plays consistently strong, close-to-optimal moves.

10. At approximately how many training episodes does the agent begin making intelligent decisions?

Ans: Noticeably intelligent play, such as reliably blocking the opponent and favouring the center, begins to appear around 500 to 1,000 training episodes.

11. At what stage does the learning appear to converge?

Ans: Performance largely stabilizes between roughly 5,000 and 10,000 episodes, after which further training produces only marginal gains in win rate.

12. Why does the win percentage improve with additional training?

Ans: With more games played, the agent's value estimates for each board state become more accurate, since they are averaged over a larger number of outcomes. This lets it more reliably distinguish good moves from bad ones, which raises its win percentage.

13. Why do most games end in a draw after sufficient training?

Ans: Once both sides of the self-play have learned strong, close-to-optimal policies, neither can be tricked into a losing blunder. Tic-Tac-Toe is a solved game where optimal play from both sides leads to a draw, so well-trained agents converge to this outcome.

14. What happens if the number of training episodes is too small?

Ans: With too few episodes the agent has not seen enough states or outcomes to form reliable value estimates. It behaves close to randomly, fails to recognize threats or opportunities, and loses far more often than a well-trained agent.

Observation

As training episodes increased, the agent's performance improved from essentially random play at 100 episodes to near-optimal, hard-to-beat play at 10,000 episodes. It progressively learned to block threats, favour strong opening squares such as the center and corners, and avoid careless mistakes, with performance leveling off once its value function had converged.

Conclusion

This lab showed how a Reinforcement Learning agent can learn to play Tic-Tac-Toe purely through self-play, without any labelled training data. Early in training the agent played almost randomly and lost frequently, but as the number of training episodes increased, it steadily learned to block threats, prefer strong positions such as the center and corners, and avoid careless mistakes. Its win rate rose and its loss rate fell sharply with more

episodes, and beyond a few thousand episodes its performance converged, with most games ending in a draw once both sides played near-optimally. This demonstrates the core RL idea that an agent can discover an effective strategy purely from trial-and-error interaction with its environment, guided only by the rewards it receives.

Screenshots of the RL Tic-Tac-Toe Demo

Episodes: 100

Reinforcement Learning Tic Tac Toe with Value Function

Demo

Train Agent

Number of Episodes

100

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

Play against agent (Skill level: 100)

Agent trained and moved. It's your turn.

Agent's Value Function

46.2%	48.4%	48.8%
48.8%	49.4%	48.8%
55.6%	48.8%	49.6%

SIMULATION

RESET GAME

Episodes: 500

Reinforcement Learning Tic Tac Toe with Value Function

Demo

Train Agent

Number of Episodes

500

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

RESET GAME

Play against agent (Skill level: 500)

Agent trained and moved. It's your turn.

Agent's Value Function

42.2%	43.1%	48.3%
49.7%	48.8%	54.8%
50.2%	48.7%	48.7%

Episodes: 1,000

Reinforcement Learning Tic Tac Toe with Value Function

Demo

Train Agent

Number of Episodes

1000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

RESET GAME

Play against agent (Skill level: 1000)

Agent trained and moved. It's your turn.

Agent's Value Function

39.6%	49.7%	50.7%
44.3%	60.3%	51.1%
49.4%	50.7%	48.2%

Episodes: 5,000

Demo

Train Agent

Number of Episodes

5000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 5000)

Agent trained and moved. It's your turn.



Agent's Value Function

47.6%	55.4%	45.6%
42.9%	87.2%	51.1%
45.3%	51.5%	50.4%

RESET GAME

Episodes: 10,000

Demo

Train Agent

Number of Episodes

10000

Number of times agent play against each other in simulation

Agent 1 Explore Probability

0.05

Agent 1 Learning Rate

0.5

Agent 2 Explore Probability

0.05

Agent 2 Learning Rate

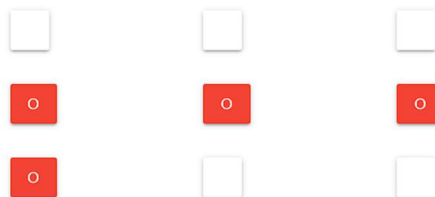
0.5

TRAIN AGENTS

SIMULATION

Play against agent (Skill level: 10000)

Agent trained and moved. It's your turn.



Agent's Value Function

45.7%	49.5%	63.5%
66.2%	91.5%	34.1%
29.9%	67.9%	46.6%

RESET GAME